NEURIPS 2026 · LP4FM WORKSHOP · POSTER

ReconSpan: Reconstruction-Guided Adaptive Latent Tokenization

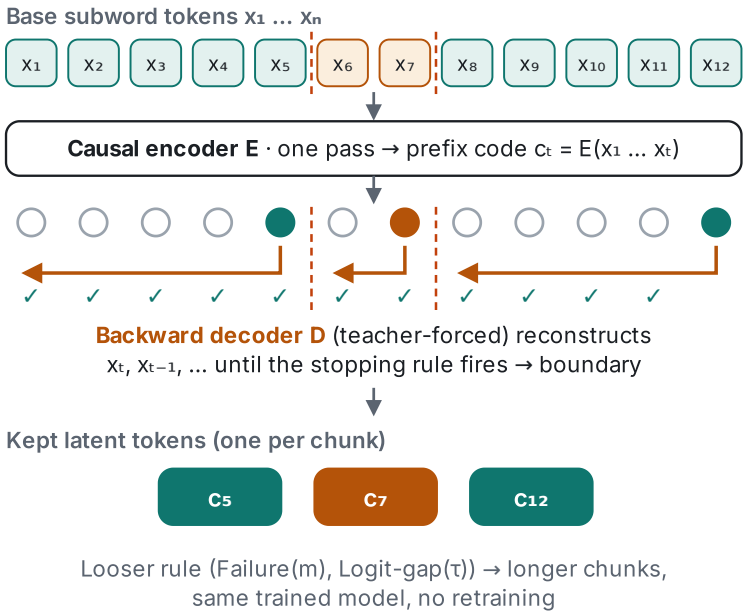
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1 PROBLEM

- Subword tokenizers fix units before the model runs, from corpus frequency: granularity ignores the input.
- Adaptive latent tokenization gives each input-dependent span one continuous latent token, so the chunking rule decides where representation and compute go.

Idea: place each boundary where a decoder can no longer reconstruct the span from a single code.

2 METHOD



Schematic (illustrative chunk sizes). Encoder: Pythia-410M; decoder: Mamba2-130M. Trained only on reconstruction; difficult spans get shorter chunks.

3 KEY RESULTS

6.5–12.2

mean base tokens per latent token from **one** trained autoencoder, set by the stopping rule

90.7%

of tokens in perfectly reconstructed chunks at matched latent count (equal spacing 81.6%, surprisal 61.7%, random 61.8%)

74.3% vs 10.0%

Llama-3-8B reading latents: AG News topic (raw text 72.6%) vs LAMBADA exact word (round trip 72.5%)